PROJECT REPORT

This is a summary of the WiDS project that I did over the last three weeks. It is called "Machine Learning from Scratch." I was inclined to do the project because I was always very excited to know more about machine learning and how AI truly works.

The good thing about this was we initially started with an introduction to what these jargon terms actually mean. Everybody keeps talking about AI, machine learning, and deep learning, without properly describing the terms, so reading about it was enlightening. Initially, we were given articles to read, which allowed us to explore what these terms exactly mean. From there, I developed a pretty intuitive understanding of what these terms actually encompass.

We then moved to a video series that further helped us document what all comes under AI, such as how to classify the different parts of AI, the terminologies, what machine learning can do, what it cannot do, and where we currently are in the stage of GPTs and other AI systems that are being developed now.

From there, we switched to a basic revision of linear algebra and probability, which was also fun because it was a good recap of what I did in my JEE days. After that, through the assignment itself, we were made to explore linear regression. Interestingly, in my first semester of studying IEOR, we did a certain section on statistics. For me, it was a really good learning experience to extrapolate that to more examples and try to apply what I had learned about linear regression into further exploring its application in ML algorithms, specifically learning more about the mathematical aspect of it.

I enjoyed reading topics through the book "ISLR." After that, when we moved to understanding how ML algorithms work, for me, understanding what regression actually refers to was a big clarity. I had seen the term thrown around a lot, but I had never actually understood what it means. A further realization was the slight difference between quantitative and qualitative ways of sorting data. There is a real difference in what we can classify into datasets and what we cannot, such as what we can put in numbers and what really has to go as a probability. For instance, a diagnosis goes under a yes or no classification, while an example we covered was the salaries of baseball hitters, depends on numerical metrics. This was good clarity to have!

One of the most interesting topics for me so far has been ensemble methods and understanding how the different modulations work when you start training an ML algorithm. I think I could explore that more on my own because it came under a lot of theoretical basis that provided me with a lot of conceptual clarity. There is an interesting simulation that I came across that helped me understand random forests, boosting, and bagging a bit more. A link to that is attached in the context. https://gemini.google.com/share/8de9d42681bd

Decision trees were also really fun to work with, as was the confusion matrix. The confusion matrix is used to help diagnosis is something that we had studied in our first semister biology course. Studying it now in a more official manner, where I calculate all the probabilities and provide reasoning as to why certain metrics are more important compared to others, felt like a true application of background knowledge. I found it very cool that we can connect something in statistics to something important by applying it in the real world.

My experience of the course so far has been great, and I wish to learn even more interesting things in the weeks ahead. Really looking forward to it! Thank you.